

🚒 LABLAB.AI • ASSEMBLYAI VOICE AGENT HACKATHON

RescueSignal AI

When communication fails, your voice becomes a signal — and when silence is survival, your touch saves lives.

Universal-1 Engine

State-of-the-art voice agent transcribing distressed speech and faint whispers in extreme acoustic debris.

Dual-Bridge Uplink

Physical FSK soundwaves through solid concrete + Disaster Tactical Wi-Fi bubbles (Starlink / Drones).

911 / 112 AI Dispatcher

Autonomous medical triage, FFT frequency spectrum monitoring, and instant ICU / SWAT team deployment.

Team: RescueSignal AI • Live URL: rescuesignal-ai.onrender.com • AssemblyAI Universal-1 • LeMUR Voice Triage

When Traditional Networks Collapse

Cellular towers, fiber backbones, and internet fail when victims need them most.

4G/5G: ✕ • INTERNET: ✕



Natural & Structural Catastrophes

Earthquakes, building collapses, explosions, and floods. Victims are pinned beneath tons of rubble and dust with zero touchscreen visibility.

- ▶ Hands pinned: cannot type or hold phone
- ▶ Voice is the only remaining lifeline
- ▶ Acoustic echo and dust choke communications



Hostile Threats & Kidnappings

Active armed intruders, abductions, and covert hostage situations. Victims are hiding in closets or being pursued by armed assailants.

- ▶ Speaking out loud is immediately fatal
- ▶ Victims can only whisper at low decibels
- ▶ Zero time: every second of delay costs lives



The Paradox of Modern Phones

Victims hold advanced smartphones with computational power, speakers, and microphones—yet they are completely silenced because cloud APIs require an active internet connection.

- ▶ Standard AI voice assistants fail completely offline
- ▶ First responders arrive blind to victim vital stats
- ▶ FIFO queuing wastes critical Golden Hour minutes

Powered by AssemblyAI Universal-1 & LeMUR

Transforming distorted spoken distress into structured, actionable life-saving orders.

Speech-to-Text & Acoustic Resilience

AssemblyAI Universal-1 was selected for its unmatched ability to decipher low-decibel whispered speech and loud distressed cries in chaotic environments.

Universal-1 Engine Automatic Language Detection Low-Latency Streaming

Emergency Domain Word Boost:

"rubble" "collapse" "unconscious" "asphyxia" "kidnap" "hostage" "broken leg"

Autonomous Medical & Tactical Reasoning (LeMUR)

RescueSignal AI does not stop at speech transcription. Its intelligent agent reasons over the transcript to extract structured triage entities:



- Semantic compression into a 39-byte RS1 packet
- 16-bit CRC checksum guarantees zero data corruption
- Immediate classification: P1 Critical vs P2 Urgent

How It Connects with AssemblyAI in Disaster Zones

DUAL-PATH UPLINK

A dual-path hybrid architecture solving the complete offline dilemma.

PATH A : The Acoustic FSK Bridge (0% Radio Network)

When a victim is buried under 3 meters of reinforced concrete where radio waves cannot penetrate:

- ▶ **On-Device RS1 Packet:** Compresses 140+ bytes into 39 bytes.
- ▶ **Acoustic FSK Modulation:** Plays 1200 / 2200 Hz audio frequencies through phone speaker; sound physically vibrates through concrete & rubble.
- ▶ **Surface Gateway Relay:** Rescuer geophones pick up the FSK signal, decode the RS1 packet, and bridge it directly to AssemblyAI Cloud via Starlink.

PATH B : Disaster Tactical Wi-Fi Bubbles

First responders deploy temporary satellite-backed wireless bubbles across the disaster perimeter:

- ▶ **Mobile Command Trucks & Drones:** Project high-power open Wi-Fi hotspots connected to Starlink satellites.
- ▶ **Instant Automatic Handshake:** Victim phone detects emergency SSID and switches to Tier 1 High-Fidelity Streaming.
- ▶ **Direct Stream to AssemblyAI Universal-1:** Raw audio streams to cloud for deep transcription, multilingual auto-translation, and LeMUR triage.

911 / 112 AI Crisis Dispatcher: Dynamic Vital Triage

Overriding First-In First-Out (FIFO) queues to save lives based on vital urgency.

#1 HIGHEST PRIORITY

P1 - CRITICAL

Sector 4 — Collapsed Rubble

Victim 1: 3 people trapped, 1 broken leg, Sarah unconscious under dust.

- ▶ Automated ICU Resuscitation Unit
- ▶ Heavy Extrication & Jaws Dispatched
- ▶ Status: 🚒 Units Dispatched (ETA: 4m)

#2 PRIORITY

P1 - TACTICAL

Sector 9 — Covert Closet

Victim 2: Whispered call: armed intruder, 2 hostages hiding in locked closet.

- ▶ SWAT Hostage Rescue Alerted
- ▶ Zero-Siren / Zero-Light Protocol
- ▶ P2P Silent Mesh Triangulation

#3 PRIORITY

P2 - URGENT

Sector 2 — Elevator Shaft

Victim 3: 4 trapped after quake, mild asthma panic, all conscious and stable.

- ▶ Shaft Winch Team Dispatched
- ▶ Secondary Paramedic Assessment
- ▶ Does NOT divert trauma ICUs from Sec 4

Technical Stack & Life-Saving Impact

Engineered for instant real-world search and rescue deployment.

Production Technology Stack

- ▶ **Voice Agent Core:** AssemblyAI Universal-1 STT SDK, LeMUR Entity Extraction, Emergency Word Boost.
- ▶ **Backend Engine:** Python 3.13, FastAPI asynchronous REST API, WebSockets P2P Mesh Relay.
- ▶ **Acoustic Synthesizer:** Pure Web Audio API FSK (1200/2200 Hz), Optical Morse Strobe, FFT Spectrogram.
- ▶ **Edge Resilience:** Zero-dependency on-device fallback parser, CRC16 error check.

Verified Submission Links

- ▶ **Live Deployment:** rescuesignal-ai.onrender.com (Render Cloud)
- ▶ **GitHub Repository:** [chaibi-mustapha/rescuesignal-ai](https://github.com/chaibi-mustapha/rescuesignal-ai)
- ▶ **HD Video Demonstration:** 1920x1080 60 FPS Studio Demo (< 3 Min, Synchronized 3-Victim Audio)
- ▶ **Bilingual Ready:** Full English & French V FR production editions